Shaan Pakala

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Google Scholar, LinkedIn & GitHub

About

I am a 1^{st} year Computer Science Ph.D. student at the University of California, Riverside, where I work on AI4Science research problems with Professor Vagelis Papalexakis. This includes developing interpretable surrogate models for designing new materials, or efficiently generating expensive physics simulation data using generative machine learning.

Education

University of California, Riverside

- **Ph.D. in Computer Science** — **Since Sept. 2025**
  Advised by Professor Vagelis Papalexakis
- **B.S. in Data Science** — **Sept. 2021 – June 2025**

Research Experience

Lawrence Livermore National Laboratory

- **Graduate Research Intern** — **Summer 2026**
  Intern in predictive healthcare group, with Mary Silva & Priyadip Ray
- **Graduate Research Intern** — **Summer 2025**
  Intern in predictive healthcare group, with Braden Soper & Priyadip Ray

University of California, Riverside

- **Graduate Student Researcher** — **Since June 2025**
  AI4Science research with Prof. Papalexakis
- **Undergraduate Student Researcher** — **June 2024 – June 2025**
  Research with Prof. Papalexakis

Awards

- **Travel Award** — 2025
  NeurIPS AI for Accelerated Material Design Workshop
- **Undergraduate Research Spotlight Award** — 2025
  Bourns College of Engineering – University of California, Riverside
- **Student Travel Award** — 2024
  IEEE International Conference on Big Data
- **NSF Research Experience for Undergraduates** — 2024
  University of California, Riverside

Papers

Conference

[1] **Tensor Methods: A Unified and Interpretable Approach for Material Design** 2026
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
ACM SIGKDD (AI for Sciences track) 2026 [PDF] [Code]

[2] **Automating Data Science Pipelines with Tensor Completion** 2024
Shaan Pakala, B. Graw, D. Ahn, T. Dinh, M.T. Mahin, V. Tsotras, J. Chen, and E.E. Papalexakis
IEEE International Conference on Big Data 2024 [Link] [PDF] [Code]

Workshop

[3] **Surrogate Modeling for the Design of Optimal Lattice Structures using Tensor Completion** 2025
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
NeurIPS AI for Accelerated Material Discovery Workshop 2025 [PDF] [Code]

[4] **Efficiently Generating Multidimensional Calorimeter Data with Tensor Decomposition Parameterization** 2025
P. Goulart*, Shaan Pakala*, and E.E. Papalexakis
ICCV Representation Learning with Very Limited Resources Workshop 2025 [PDF] [Code]

[5] **Tensor Completion for Surrogate Modeling of Material Property Prediction** 2025
Shaan Pakala, D. Ahn, and E.E. Papalexakis
AAAI Bridge on Knowledge-Guided Machine Learning 2025 [PDF]

** denotes equal contribution*